

Minkowski Distance Based Pilot Protection for Tie Lines Between Offshore Wind Farms and MMC

Zhe Yang , Member, IEEE, Ruijin Zhu , and Wenlong Liao , Member, IEEE

Abstract—Offshore wind farms (OWFs) with modular multilevel converter high-voltage dc (MMC-HVdc) have become an important form of renewable energy utilization. However, if a fault occurs at the tie line between the MMC and the OWF, the fault steady-state current at the fault point will be equal to zero when the negative-sequence current is suppressed, so traditional differential protection may fail to operate. To cope with this issue, this article proposes a new pilot protection method based on Minkowski distance. For internal faults, OWF and MMC will have different transient currents, so the Minkowski distance will be much higher than 0, but it will be equal to 0 for external faults and normal operation since the fault currents on both sides are the opposite. Therefore, the internal fault can be detected reliably. The proposed method does not depend on the power frequency phasor extraction, so it is not affected by the frequency offset in the transient process. In addition, this method operates quickly and has a strong ability to withstand fault resistance and environmental noise. Moreover, since there is always a transient process when a fault occurs, the proposed method applies to different fault ride-through strategies. PSCAD simulation and real-time digital simulator experiments show that the proposed method is suitable for different fault locations and types.

Index Terms—Minkowski distance, offshore wind farms (OWFs), pilot protection, transient current.

I. INTRODUCTION

OFFSHORE wind farms (OWFs) have been built around the world due to the advantages of high utilization hours of power generation and less environmental pollution [1], [2]. They

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Zhe Yang is with the Department of Electrical Engineering, The Hong Kong Polytechnic University, Hong Kong (e-mail: zhet1yang@polyu.edu.hk).

Ruijin Zhu is with the School of Electrical Engineering, Tibet Agriculture and Animal Husbandry University, Nyingchi 860000, China (e-mail: zhuruijin@xza.edu.cn).

Wenlong Liao is with Wind Engineering and Renewable Energy Laboratory, École Polytechnique Fédérale de Lausanne, 1015 Lausanne, Switzerland (e-mail: wenlong.liao@epfl.ch).

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are often connected to onshore power systems via modular multilevel converter high-voltage dc (MMC-HVdc) technologies due to their good transmission capabilities [3]. Both sides of the tie line between the OWF and the MMC are power converters, so unique fault behaviors will be produced due to different control strategies, which may seriously threaten the correct operation of traditional protection [4], [5].

Differential protection is usually used as the main protection of the transmission line due to its absolute selectivity and fast operation speed [6]. At present, the research on the impact of wind farm access on differential protection mainly focuses on the lines between renewable energy sources (RESs) and the traditional grid. Prasad et al. [7] pointed out that conventional proportional restraint differential protection can still operate correctly when the power grid is strong. However, when the short-circuit capacity of RESs reaches one-quarter of that of the connected grid, the operating performance of the traditional differential protection will decline or even fail to operate [8]. To solve this problem, many new pilot protection methods have been proposed. Bolandi et al. [9] proposed the concept of differential impedance, but the voltage and current information must be used simultaneously, and it has high requirements for the synchronization of the signals on both sides. In addition, some scholars have proposed a differential protection method based on the current amplitude ratio, but the amplitude of the fault current on both sides may not have evident differences for the OWF-MMC system [10]. In [11], a pattern recognition-based pilot protection method was proposed. Since only the line characteristics were utilized, it is suitable for different backside power sources. However, this method uses differential calculations, so the ability to tolerate noise is limited. In addition, a pilot protection method based on traveling waves was proposed in [12] and [13], and it is also applicable to different power sources, but accurate wavefront detection is still difficult, and these methods have high requirements for sampling rates. Meanwhile, the fault current of wind farms is always different from that of the synchronous generators, so this difference was evaluated by Pearson correlation coefficient, Cosine similarity, and Euclidean distance in [14], [15], and [16] to form new pilot protection methods. However, the applicability of these methods to the OWF-MMC system has yet to be tested.

Some scholars have also carried out preliminary research on the adaptability of differential protection on the OWF-MMC

system. Zheng et al. [17] illustrated that MMC could be blocked due to overcurrent when traditional voltage-frequency control was used, so traditional differential protection will fail to operate because of the short fault data window. Accordingly, a pilot protection scheme based on singular value decomposition was proposed. Similarly, distance protection was also challenged due to the short data window, so a distance protection scheme based on parameter identification was suggested in [18]. However, many wind farms may be connected to the same MMC station, so all the wind farms will be out of service due to the blocking of the MMC. In this way, a huge power shortage will threaten the stability of the onshore power system. Gao et al. [19] proposed that MMC should use a grid-forming controller to limit the output fault current during a fault so that MMC will not be blocked due to overcurrent. On this basis, the adaptability of traditional differential protection for three-phase faults was analyzed, and a new active control-based method was proposed to improve the performance of differential protection by reducing the angle difference between both sides. However, this method is only suitable for three-phase faults, not for asymmetrical faults. Shi et al. [20] pointed out that the steady-state current at the fault point is close to zero under single-phase grounding faults and phase-to-phase faults for the OWF-MMC system when MMC adopts the grid-forming controller. However, the impact of this unique fault behavior on traditional differential protection is not analyzed in this literature.

Therefore, existing methods are either not suitable for the studied system with the MMC adopting the grid-forming control or cannot be applied to all fault types. To cope with this issue, a new pilot protection method based on the Minkowski distance is proposed. The main contributions of this article are as follows.

- 1) Combined with the unique fault behaviors of the OWF-MMC system, it is found that traditional differential protection has poor performance and even refuses to operate.
- 2) The proposed pilot protection method based on the Minkowski distance can operate quickly and is suitable for different fault types and different fault ride-through (FRT) strategies.
- 3) The proposed method has a strong ability to withstand fault resistance and noise interference.

The rest of this article is organized as follows. Section II illustrates the studied system topology and the used controller of the OWF and MMC. Under this circumstance, the fault behaviors of the OWF-MMC system will be clarified and the adaptive analysis for conventional differential protection is done in Section III. To deal with this issue, a Minkowski distance-based pilot protection with the corresponding setting rules is proposed in Section IV. After that, Extensive simulations in PSCAD and real-time digital simulator are performed to confirm the proposed method in Sections V and VI. Finally, Section VII concludes this article.

II. STUDIED SYSTEM

The tested system in this article is shown in Fig. 1. OWFs are connected to onshore systems via the MMC-HVdc line. The wind farm is composed of permanent magnet synchronous

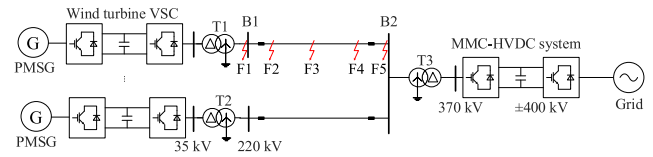


Fig. 1. Tested OWF-MMC system.

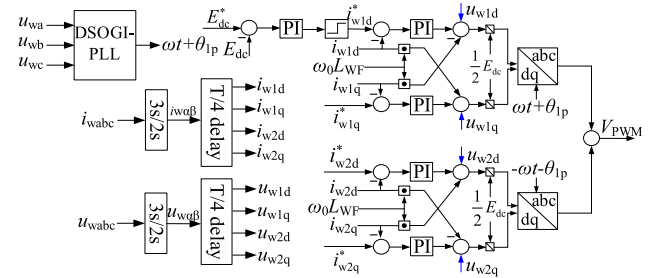


Fig. 2. Control system of the OWF.

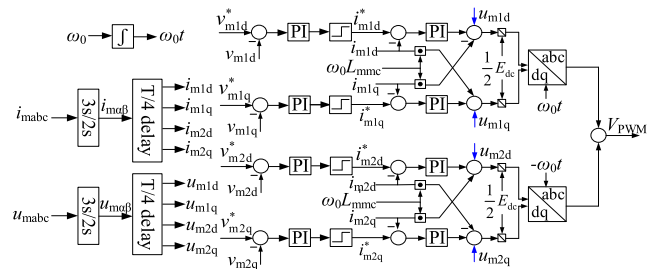


Fig. 3. Control system of the MMC.

generator (PMSGs)-based wind turbines, and they are connected to the tie line through back-to-back converters. Five fault points are set on the tie line: two external faults F1 and F5 are set on Bus B1 and B2, respectively, and three internal faults F2, F3, and F4 are set at 0%, 50%, and 100% of the tie line away from B1.

The fault characteristics of PMSGs are determined by the control system of the grid-side voltage-source converter, so Fig. 2 shows the block diagram of its control system. The grid-side controller obtains the phase-locked angle through the double second-order generalized integrator phase-locked loop, and the positive- and negative-sequence separation is realized through a 1/4 cycle delay method [21]. During normal operation, the positive-sequence active current reference of the current inner loop is given by the dc voltage outer loop, and both the positive-sequence reactive current reference and negative-sequence current references are set to 0. Once a fault occurs, the dc voltage outer loop will be cut off, and the positive sequence current references are directly determined according to the grid code, and negative sequence current references are still set to 0.

The MMC will adopt the grid-forming controller with positive- and negative-sequence separation, as shown in Fig. 3. For this controller, the phase-locked angle no longer comes from the PLL but is obtained by the integral for the angular

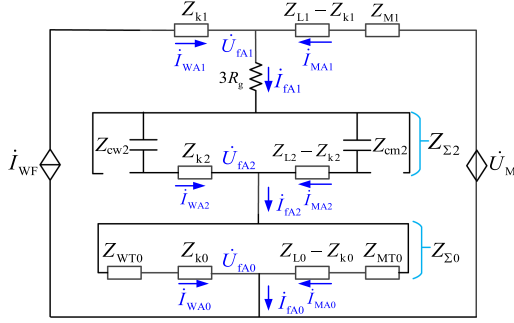


Fig. 4. Sequence network for AG faults.

velocity of the fundamental frequency. In normal operation, the ac voltage control generates the reference values of the current inner loop [22]. To prevent overcurrent, the current limiting loop is installed at the output of the proportional-integral (PI) controller. During a fault, the positive sequence ac voltage loop still works, but the outer loop of negative sequence voltage is cut off. After that, negative-sequence current references are directly set to 0.

III. PROBLEM STATEMENT

In this part, the fault characteristics of the OWF-MMC system will be analyzed under different fault types, and the adaptive analysis of traditional proportional braking differential protection will be done. Due to the rotational symmetry in the fault analysis, line-A (AG) ground faults, line-B-to-line-C (BC) faults, and line-B-to-line-C ground (BCG) faults are respectively taken as examples to represent single-line ground faults, line-to-line faults, and line-to-line ground faults.

A. Single-Line Ground Faults

When the control system shown in Section II is adopted, both the MMC and the OWF will eliminate the negative sequence currents during a fault. Under this circumstance, the sequence network for AG faults is depicted in Fig. 4. In the figure, Z_k is the line impedance from the fault point to the OWF-side relay point, Z_L denotes the total line impedance, Z_M is the sum of the impedances of the MMC-side transformer and the outlet filter, Z_{cw} is the line ground capacitance on the OWF side, and Z_{cm} denotes the line ground capacitance on the MMC side. In addition, Z_{WT} and Z_{MT} denote the equivalent impedances of the transformers on the OWF side and MMC side, respectively. $\dot{U}_{fA}$ denotes the voltage at the fault point, and the subscripts 1, 2, and 0 represent positive-, negative-, and zero-sequence components, respectively. The wind farm is equivalent to a controlled current source in the positive sequence fault circuit $\dot{I}_{WF}$, and the MMC is equivalent to a controlled voltage source $\dot{U}_M$ since the positive-sequence voltage outer loop still works during the fault. In addition, since both the OWF and the MMC adopt a control strategy to suppress the negative-sequence current, they are equivalent to an open circuit in the negative-sequence fault circuit. At this time, the fault current can only flow through the line ground capacitance.

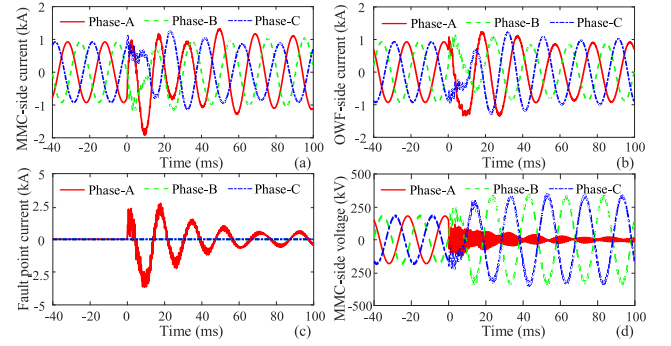


Fig. 5. Fault current at different points and voltage on the MMC side for AG faults. (a) MMC side. (b) OWF side. (c) Fault point. (d) Fault voltage.

It can be seen from the sequence network diagram that the sequence fault current at the fault point can be expressed as

$$\dot{I}_{fA1} = \dot{I}_{fA2} = \dot{I}_{fA0} = \frac{\dot{U}_{fA1}}{Z_{\Sigma 2} + Z_{\Sigma 0} + 3R_g}. \quad (1)$$

Since $Z_{\Sigma 2}$ is mainly composed of line capacitive reactance with a large amplitude, the sequence current at the fault point is very small. In this situation, there is

$$\dot{I}_{fA1} \approx -\dot{I}_{fA2}. \quad (2)$$

It can be seen that (2) the fault currents on both sides still basically maintain an opposite state. In addition, since $Z_{\Sigma 2}$ is much greater than $Z_{\Sigma 0}$ and $3R_g$, $\dot{U}_{fA2}$ will be basically equal to $\dot{U}_{fA1}$, and $\dot{U}_{fA0}$ will be close to 0, that is

$$\begin{cases} \dot{U}_{fA1} = -\dot{U}_{fA2} \\ \dot{U}_{fA0} \approx 0 \end{cases}. \quad (3)$$

In this case, the voltages of the nonfaulted phases will be equal to

$$\begin{cases} \dot{U}_{fb} = \alpha^2 \dot{U}_{fA1} + \alpha \dot{U}_{fA2} = -j\sqrt{3} \dot{U}_{fA1} \\ \dot{U}_{fc} = \alpha \dot{U}_{fA1} + \alpha^2 \dot{U}_{fA2} = j\sqrt{3} \dot{U}_{fA1} \end{cases}. \quad (4)$$

As shown in (4), the voltage for the nonfaulty phase will increase by $\sqrt{3}$ times. For an AG metallic fault at F3, the fault current at the OWF side, the MMC side, the fault point, and the fault voltage at the MMC side are shown in Fig. 5.

As seen in Fig. 5(a) and (b), the fault current on the MMC side and the OWF side gradually returns to the state of three-phase symmetry after one or two cycles of the transient process is finished, so the fault characteristics for this OWF-MMC system are not obvious. Meanwhile, the fault current at the fault point in Fig. 5(c) is suppressed after the transient process. In addition, the voltage for the nonfaulty phase is increased by $\sqrt{3}$ times, while the voltage for the faulty phase voltage is decreased to nearly 0. These phenomena are consistent with theoretical analysis.

B. Line-to-Line Faults

For BC faults, its sequence network diagram is shown in Fig. 6, where the positive-sequence circuit and the negative-sequence circuit are connected in parallel.

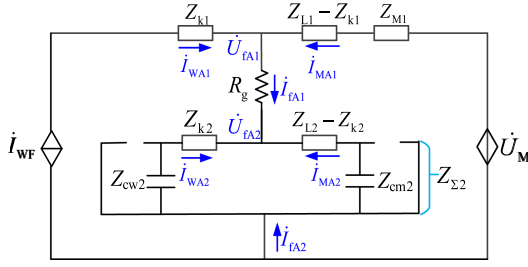


Fig. 6. Sequence network for BC faults.

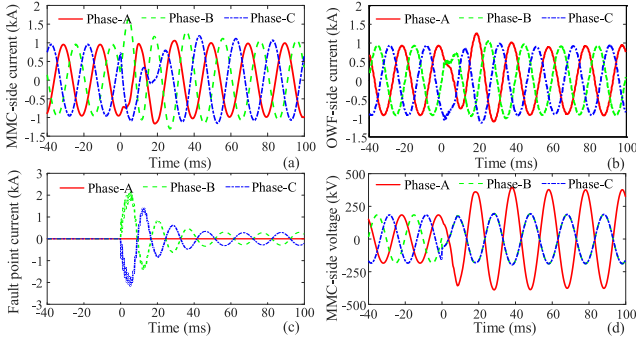


Fig. 7. Fault current at different points and voltage on the MMC side for BC faults. (a) MMC side. (b) OWF side. (c) Fault point. (d) Fault voltage.

According to the sequence network diagram in Fig. 6, the sequence fault current at the fault point can be obtained

$$\dot{I}_{fA1} = -\dot{I}_{fA2} = \frac{\dot{U}_{fA1}}{Z_{\Sigma 2} + R_g}. \quad (5)$$

Since $Z_{\Sigma 2}$ is dominated by the line ground capacitance, the positive and negative sequence fault current at the fault point is very small. Meanwhile, $Z_{\Sigma 2}$ is much larger than R_g , so there is

$$\dot{U}_{fA1} = \dot{U}_{fA2}. \quad (6)$$

The phase-A voltage during a fault is equal to the sum of $\dot{U}_{fA1}$ and $\dot{U}_{fA2}$, so the phase-A voltage at the fault point will rise to twice the rated voltage when a BC fault happens. When a BC metallic fault occurs at F4, the fault current on the OWF side, the MMC side, the fault point, and the fault voltage on the MMC side are given in Fig. 7.

After a short transient process, the fault currents on the OWF side and the MMC side in Fig. 7(a) and (b) gradually return to the state of three-phase symmetry. Similarly, only a small fault current is present at the fault point after one cycle, as shown in Fig. 7(d). Meanwhile, the voltage in Fig. 7(c) for phase-A during a fault will be increased by 2 times. In order to suppress the overvoltage for AG and BC faults, the positive-sequence voltage reference of the MCC can be reduced from 1 to 0.5 once a fault is detected.

C. Line-to-line Ground Faults

The fault sequence network diagram for BCG faults is shown in Fig. 8, where negative- and zero-sequence circuits are connected in parallel.

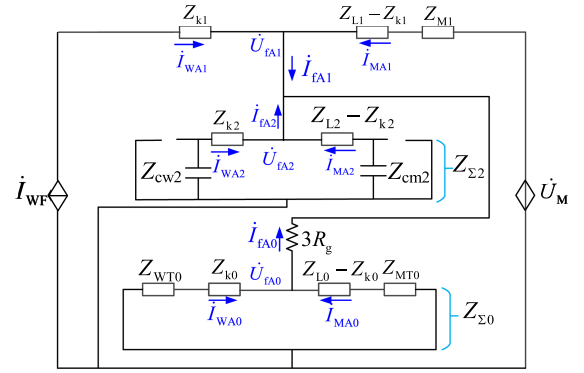


Fig. 8. Sequence network for BCG faults.

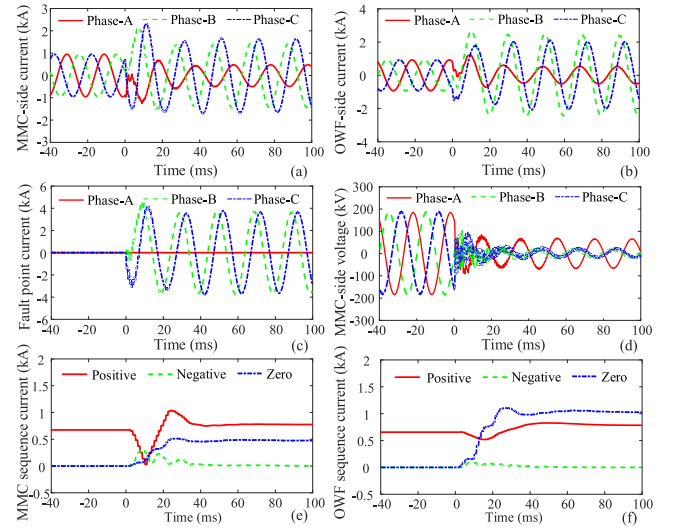


Fig. 9. Fault current at different points and voltage on the MMC side for BCG faults. (a) MMC side. (b) OWF side. (c) Fault point. (d) Fault voltage. (e) MMC-side sequence current. (f) OWF-side sequence current.

According to this diagram, the positive sequence current at the fault point can be expressed as

$$\dot{I}_{fA1} = \frac{\dot{U}_{fA1}}{Z_{\Sigma 2} \parallel (3R_g + Z_{\Sigma 0})} \approx \frac{\dot{U}_{fA1}}{(3R_g + Z_{\Sigma 0})}. \quad (7)$$

Since $Z_{\Sigma 2}$ is far higher than $(3R_g + Z_{\Sigma 0})$, most $\dot{I}_{fA1}$ can flow through the zero-sequence fault circuit such that the negative-sequence circuit can be seen as an open circuit. Therefore, the fault current for BCG faults at the fault point shows a large value. When a BCG fault occurs at F2, the fault current on the MMC side, the OWF side and the fault point, and the fault voltage on the MMC side are displayed in Fig. 9.

It can be seen from Fig. 9(a) and (b) that the fault current for phase-B and phase-C will be increased significantly. Meanwhile, the fault current at the fault point is also significantly greater than the rated current, and the voltage for the faulty phase will decline hugely, as shown in Fig. 9(c) and (d). The sequence decomposition is performed for fault currents on the OWF side and MMC side, and it is found that positive- and zero-sequence currents are dominated, and the negative-sequence current is

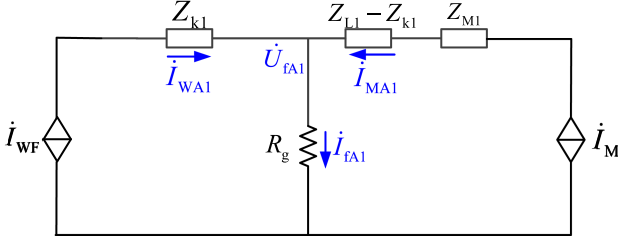


Fig. 10. Sequence network for ABCG faults.

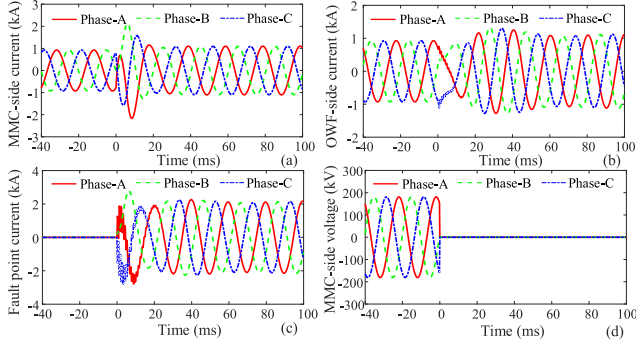


Fig. 11. Fault current at various points and voltage on the MMC side for ABCG faults. (a) MMC side. (b) OWF side. (c) Fault point. (d) Voltage.

equal to 0, as shown in Fig. 9(e) and (f). These phenomena are consistent with theoretical analysis.

D. Three-Phase Faults

For three-phase ground (ABCG) faults, the equivalent fault network is shown in Fig. 10. Since the system voltage will decrease hugely, the current limiting loop after the PI controller in the voltage control loop will work, so the MMC is also equivalent to a controlled current source.

At this time, the fault currents on both sides are determined by their current references. The analytic expression of the steady-state fault current is as follows:

$$i_{fa} = \sqrt{i_{1d}^{*2} + i_{1q}^{*2}} \sin \left(\omega t + \theta_{1p} + \arctan \left(\frac{i_{1q}^*}{i_{1d}^*} \right) + 90^\circ \right). \quad (8)$$

It should be noted that the MMC and the OWF have their own phase-locking angles, so the angle difference between both sides of the fault current is determined by the phase-locked angle difference and the current reference difference.

When an ABCG fault arises at F4, the fault currents on the OWF side, MMC side, fault point, and the fault voltage on the MMC side are shown in Fig. 11.

When an ABCG fault occurs, the MMC and OWF fault currents in Fig. 11(a) and (b) keep balanced and reach the current limiting value. Meanwhile, the fault current at the fault point presents a large value, as displayed in Fig. 11(c). Since F4 is located at the output of the relay point on the MMC side, the fault voltage on the MMC side drops to 0, as shown in Fig. 11(d).

In summary, the fault current at the fault point is very small for AG faults and BC faults, but it is still large for BCG and ABCG faults.

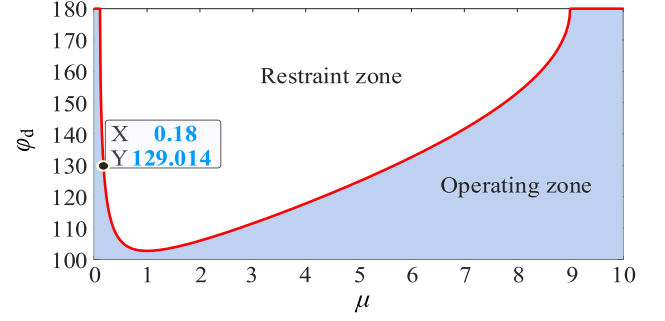


Fig. 12. Operating boundary for traditional differential protection.

E. Protection Adaptive Analysis

For traditional grids, the fault current angles on both sides during external faults and normal operation are completely opposite, but they are the same for internal faults, so the basic principle for differential protection is [23]

$$\begin{cases} I_{op} \geq I_{op0}, \text{ and } I_{op} \geq k I_{re} \\ I_{op} = |\dot{I}_W + \dot{I}_M|, I_{re} = |\dot{I}_W - \dot{I}_M| \end{cases} \quad (9)$$

where I_{op} denotes the operating current, I_{re} denotes the restraint current, $\dot{I}_W$ and $\dot{I}_M$ represent the fault currents on the wind farm side and the MCC side, and k denotes the restraint coefficient.

Assumed that μ is equal to I_M/I_W , and the angle difference between $\dot{I}_W$ and $\dot{I}_M$ is defined as φ_d . At this time, I_{op} and I_{re} can be expressed as (10) according to the Law of cosines

$$\begin{aligned} I_{op} &= I_W \sqrt{1 + \mu^2 - 2\mu \cos(180^\circ - \varphi_d)} \\ I_{re} &= I_W \sqrt{1 + \mu^2 - 2\mu \cos \varphi_d} \end{aligned} \quad (10)$$

Substitute (10) into (9), the operating boundary of differential protection can be obtained

$$\cos \varphi_d > \frac{(k^2 - 1) \mu^2 + (k^2 - 1)}{2\mu(1 + k^2)}. \quad (11)$$

According to (11), the relationship between μ and φ_d under the operating boundary can be drawn in Fig. 12.

For the OWF-MMC system, the fault current on both sides is still completely opposite for AG and BC faults, so the operating point is easy to fall into the restraint zone such that differential protection fails to operate.

IV. MINKOWSKI DISTANCE-BASED PILOT PROTECTION

In order to improve the protection performance, a new pilot protection method based on Minkowski distance is proposed in this part, and detailed setting principles are also given.

A. New Protection Principle

Minkowski distance is an efficient way to measure the space distance between two vectors, and its expression is as follows [24]:

$$L(x, y) = \left(\sum_{j=1}^n |x_j - y_j|^p \right)^{\frac{1}{p}} \quad (12)$$

where x_j and y_j are any element of the vectors x and y , respectively, and the value of p will be discussed in Section IV-B.

In the protection field, two vectors correspond to the sampling sequences of the currents on both sides respectively, and Minkowski distance can effectively measure the difference of the transient currents on both sides. During normal operation or external faults, the currents on both sides are opposite, so they are the same after the current on one side is reversed. At this time, the Minkowski distance will be equal to 0, that is

$$\begin{aligned} L(x, y) &= \left(\sum_{j=1}^n |x_j - y_j|^p \right)^{\frac{1}{p}} \\ &= \left(\sum_{j=1}^n |i_{Wj} - (-i_{Mj})|^p \right)^{\frac{1}{p}} = 0. \end{aligned} \quad (13)$$

When a fault arises, it is a big disturbance for the converter including the OWF and the MMC such that the negative-sequence DQ-axis currents cannot follow their reference values during the fault transient. Therefore, the negative-sequence current from the OWF and the MMC is no longer equal to 0 during the fault transients. In this way, the transient current at the fault point will have an evident inrush current, which also means that the fault currents on both sides will have a big difference for internal faults, so the Minkowski distance will be much higher than 0, that is

$$\begin{aligned} L(x, y) &= \left(\sum_{j=1}^n |x_j - y_j|^p \right)^{\frac{1}{p}} \\ &= \left(\sum_{j=1}^n |i_{Wj} - (-i_{Mj})|^p \right)^{\frac{1}{p}} = \left(\sum_{j=1}^n |i_{fj}|^p \right)^{\frac{1}{p}}. \end{aligned} \quad (14)$$

As seen in (14), $L(x, y)$ reflects the current feature at the fault point. Although the transient time only persists for one or two cycles, it is enough to trigger the tripping. Therefore, the protection criterion is as follows:

$$L(x, y) > L_{\text{set}} \quad (15)$$

where L_{set} is the threshold value, and it will be detailed in Section IV-B.

For the time window length to calculate Minkowski distance, if it is too short, it is not conducive to the numerical stability of the calculation results. On the other hand, fault transients last only one or two cycles, so the operation speed will be affected if a long time window is used. Combining these two factors, the time window length is chosen to be 10 ms. To sum up, the protection flowchart is given in Fig. 13.

First, the fault current on both sides is normalized by dividing the peak value of the rated current, and then they are low-pass filtered with a cut-off frequency of 200 Hz. After that, the MMC side current is reversed, and the Minkowski distance is calculated using 10 ms of the data window. The selection of the data window is to consider the compromise between protection speed and numerical stability. The shorter window can increase the

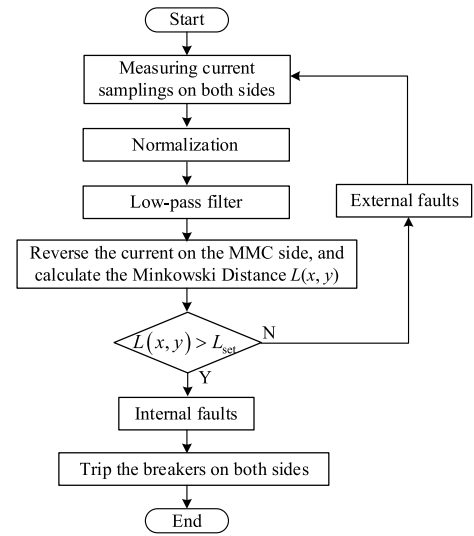


Fig. 13. Protection flowchart.

protection speed, but the calculated value will fluctuate within a large scope. For the studied system, the total fault data window can only last for 30–40 ms, so the selected data should not be too long. Considering these factors, the data window used to calculate the Minkowski distance is taken as 10 ms. If the Minkowski distance is higher than the protection setting value, an internal fault is detected; if not, an external fault is identified.

When the proposed method is applied in engineering, the communication rules should be adjusted accordingly since the 10 ms of data needs to be transmitted to the remote end instead of a single phasor.

B. Threshold Value Setting

The threshold value setting mainly considers two points: 1) the amplitude error and the phase angle error of the current on both sides and 2) the influence of the line capacitive current. The amplitude error is considered as 10%. The phase error mainly considers the sampling error and the sensing error of the current transformer (CT). The current sampling error on both sides is usually less than 5% of the sampling interval, so the angle error is less than 0.9° when the sampling rate is 5 kHz [25]. In addition, the sensing of CTs will also lead to a smaller error, and the total angle error is set to 7° . Assume that the Minkowski distance caused by the above errors is L_μ .

In addition, the ground capacitance for the cable is much larger than the overhead line, so the ground capacitive current cannot be ignored for the protection setting. According to the analysis in Section III, the biggest overvoltage for the nonfaulty phase is 2 times the rated voltage when BC faults occur. Considering this worst scenario, the capacitive current amplitude of the nonfault phase is

$$I_{c\max} = \frac{2U_N}{\sqrt{3}X_C}. \quad (16)$$

Minkowski distance of the capacitive current within 10 ms (half of the cycle) is a constant, which has nothing to do with

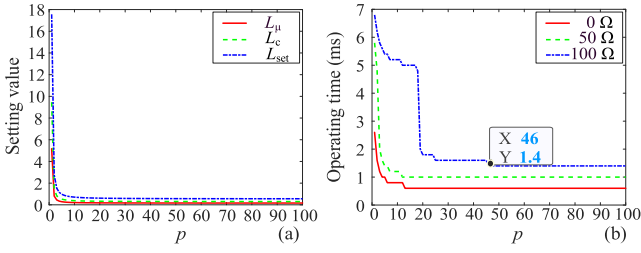


Fig. 14. Relationship between the setting value, operating time, and p . (a) L_μ , L_c , and L_{set} versus p . (b) Operating time versus p .

the initial phase and is named L_c . Based on this, a certain margin must be considered when the threshold value is set

$$L_{set} = k_{rel} (L_\mu + L_c) \quad (17)$$

where k_{rel} is the reliability coefficient, and its value ranges from 1.2 to 1.3.

The values of L_μ , L_c , and L_{set} under different p are shown in Fig. 14(a). When this setting principle is used, the relationship between the operating time and p is given in Fig. 14(b) (the simulation scenario is an AG fault at F2).

As shown in Fig. 14(a), the setting value decreases exponentially with the increase of p , so the operating time in Fig. 14(b) will also be shortened rapidly with the increase of p . When an AG fault with 100Ω of fault resistance occurs at F2, the operating time is fixed at 1.4 ms after p is increased to 46, so p is set to 46 in this article.

V. SIMULATION ANALYSIS

This section verifies the performance of the proposed method under different fault scenarios. The capacity of the OWF is 250 MW, the cable line length is 20 km, and the system frequency is 50 Hz. According to the proposed setting principle, the threshold value is 0.5735, and the sampling frequency of the protective relay is 5 kHz. Other parameters are shown in Table IV in Appendix. All the faults occur at 0 s, and the state before 0 s is also provided in the simulation results to compare the response of the proposed algorithm before and during a fault. During normal operation, the OWF keeps the unity power factor, and the Germany FRT strategy in [26] is implemented during a fault.

A. Adaptive Analysis of Differential Protection

When an AG fault with 100Ω of R_g occurs at F2, the fault currents on the MMC side and OWF side, and the fault current at the fault point are shown in Fig. 15.

After experiencing a very short transient process, the phase-A fault current on both sides in Fig. 15(a) gradually returns to the opposite state. For phase B and phase C (nonfaulty phases), the currents on both sides in Fig. 15(b) and (c) are always opposite. Since the control strategy of suppressing the negative sequence current is adopted on both sides, the fault current at the fault point in Fig. 15(d) is also suppressed. An independent measurement point measures the current at the fault point, and the protective

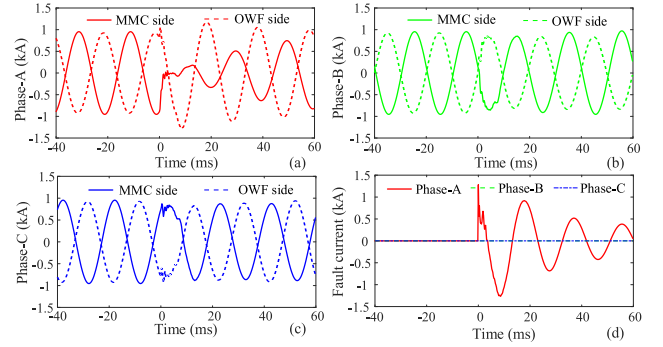


Fig. 15. Fault currents on both sides and fault point. (a) Phase-A. (b) Phase-B. (c) Phase-C. (d) Fault point.

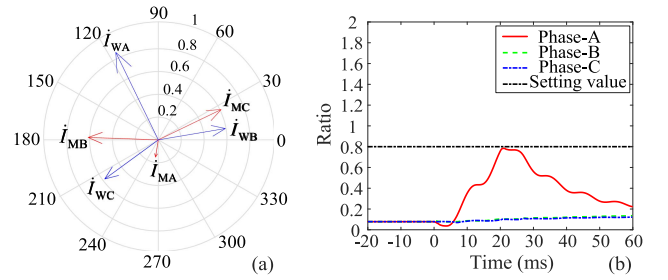


Fig. 16. Current phasor diagram and the performance of differential protection. (a) Current phasor diagram. (b) Differential protection.

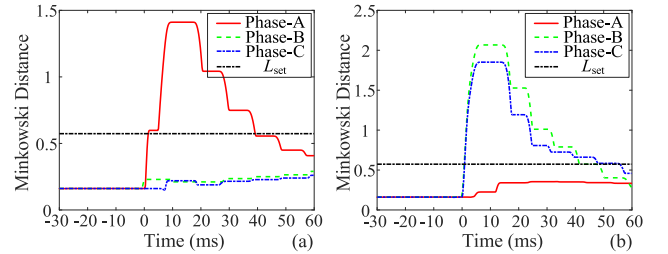


Fig. 17. Performance of the proposed method. (a) AG. (b) BC.

relays will only utilize the fault current data at the relay point on both sides.

At this time, the current phasor diagram and the performance of the differential protection at 20 ms after the fault time are given in Fig. 16. The restraint coefficient is set to 0.8 according to [14].

In Fig. 16(a), μ is equal to 0.18, and φ_d is equal to 144.975° for phase A, so differential protection will fail to operate according to the operating boundary in Fig. 12. Therefore, the ratio of the operating current to the restraint current for phase A is lower than the setting value, as shown in Fig. 16(b).

B. Different Fault Types and Fault Locations

The proposed method is tested in the same fault situation, the result is shown in Fig. 17. Meanwhile, the performance for a BC fault with 20Ω of R_g is also given in Fig. 17.

TABLE I
PERFORMANCE UNDER VARIOUS FAULT POINTS AND TYPES

Fault point	Fault type	Phase-A	Phase-B	Phase-C
F1	AG	0.207	0.272	0.189
	BC	0.230	0.110	0.121
	BCG	0.329	0.086	0.155
	ABCG	0.167	0.059	0.116
F2	AG	3.407	0.269	0.189
	BC	0.230	2.259	2.040
	BCG	0.329	4.661	3.400
	ABCG	2.591	2.827	2.230
F3	AG	3.396	0.287	0.194
	BC	0.237	2.561	2.336
	BCG	0.331	4.658	3.624
	ABCG	2.665	3.021	2.667
F4	AG	3.501	0.291	0.204
	BC	0.248	2.399	2.165
	BCG	0.347	4.787	3.705
	ABCG	2.666	2.998	2.456
F5	AG	0.237	0.292	0.204
	BC	0.234	0.116	0.118
	BCG	0.347	0.076	0.184
	ABCG	0.192	0.052	0.148

The bold value means the calculated value is higher than the threshold value.

It can be seen from Fig. 17(a) that the computed Minkowski distance for phase A exceeds the setting value from 1.6 to 39 ms, so the protection can operate quickly, while it is always lower than the protection setting value for the nonfaulty phase, so the proposed method can block the nonfaulty phase reliably. For BC faults, the situation is similar, and the operating time for phase B and phase C exceed the setting value at 0.8 ms and 1 ms, respectively, as shown in Fig. 17(b).

For different fault types and fault locations, the performance of the proposed protection is shown in Table I, and all the values are from 10 ms after the fault time.

Minkowski distance in Table I for the faulty phase is always greater than the protection setting value. However, the transient current comes from the uncontrolled process of the OWF and the MMC, so the calculated Minkowski distance does not show a certain regularity for different fault locations. Meanwhile, it is always lower than the setting value for the nonfaulty phase, so the proposed protection scheme can operate correctly for internal faults. For external faults, since the fault currents on both sides are always opposite, the calculated Minkowski distance for three phases is always lower than the threshold value. Therefore, the proposed protection can be blocked.

C. Different FRT Strategies

Besides the FRT strategy in [26], the operation performance of the proposed protection under the Danish and North American FRT is also validated. When AG and BC metallic faults are set at F3, the computed Minkowski distance for three phases is displayed in Fig. 18.

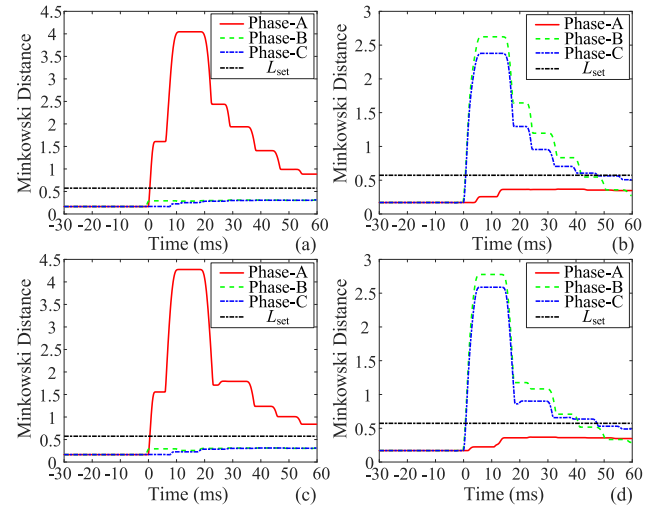


Fig. 18. Performance of the proposed method under different FRT strategies. (a) AG, Denmark. (b) BC, Denmark. (c) AG, North America. (d) BC, North America.

TABLE II
PERFORMANCE UNDER DIFFERENT FAULT RESISTANCES

Fault type	Fault resistance (Ω)	Minkowski Distance		
		Phase-A	Phase-B	Phase-C
AG	25	2.864	0.265	0.246
	50	2.230	0.253	0.253
	75	1.783	0.245	0.255
	100	1.460	0.255	0.262
BCG	25	0.270	3.564	2.587
	50	0.235	2.768	2.538
	75	0.215	2.142	1.956
	100	0.203	1.726	1.559

The bold value means the calculated value is higher than the threshold value.

As known in Fig. 18, no matter whether the Danish strategy or the North American FRT strategy is adopted, the computed Minkowski distance for the faulty phase can quickly exceed the threshold value, while it will always be lower than the protection setting value for the nonfaulty phase. Therefore, this protection can operate correctly.

D. Different Fault Resistances

For ground faults, 100 Ω is enough to cover most scenarios [27], so the fault resistances are set to 25, 50, 75, and 100 Ω , respectively. When the AG and BCG faults occur at F2, the calculated Minkowski distance is shown in Table II. All values are from 10 ms after the fault time.

As the fault resistance increases, the difference between the transient currents on both sides will be decreased, so the value of the Minkowski distance will decline. However, it can be seen from Table II that the Minkowski distance for the faulty phase is still higher than the setting value when the fault resistance is increased to 100 Ω for AG and BC faults, so the suggested protection scheme can operate correctly.

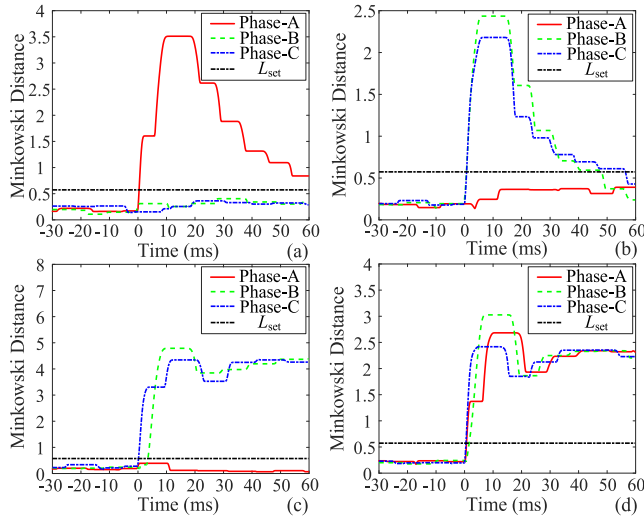


Fig. 19. Performance of the proposed method under an SNR of 20 dB. (a) AG. (b) BC. (c) BCG. (d) ABCG.

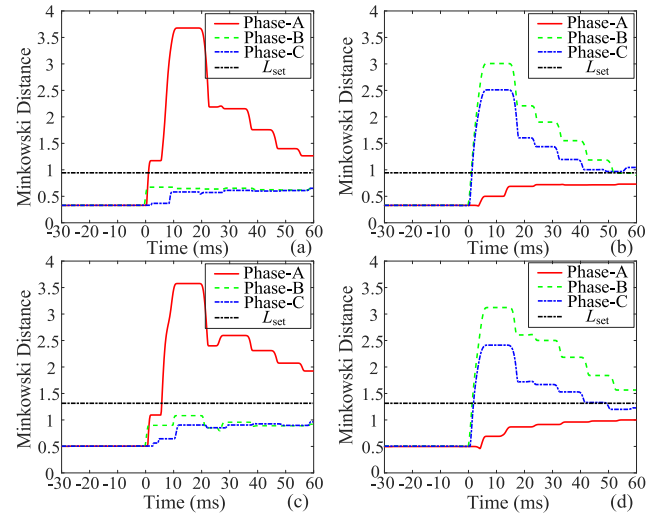


Fig. 21. Performance of the proposed method for different line lengths. (a) 40 km, AG. (b) 40 km, BC. (c) 60 km, AG. (d) 60 km, BC.

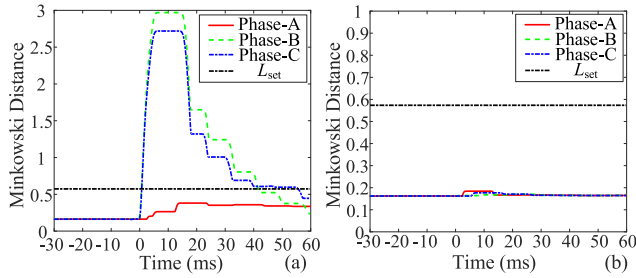


Fig. 20. Operation of the proposed protection under different cases. (a) Case 1. (b) Case 2.

E. Different Noise Interference

In this section, the impact of noise interference on the proposed method will be tested. The maximum signal-to-noise ratio (SNR) is usually 25 dB for cable lines [16]. When the SNR is taken as 20 dB, the computed Minkowski distance is shown in Fig. 19 for different fault types at F4. The added noise is Gaussian white noise.

The noise will lead to the difference between both sides of currents during normal operation, so the computed Minkowski distance before the fault in Fig. 19 will be increased. However, the low-pass filter will suppress the impact of the noise, so the Minkowski distance before the fault is still lower than the threshold value, so the protection can be blocked reliably. Similarly, the nonfaulty phase during a fault can be also blocked.

F. Impact of the OWF Power Output

Two cases will be tested. 1) the power output of the OWF only reaches 80% of the nominal current in normal operation, and a BC fault arises at F2 at 0 s. 2) The power output will be changed from 0.8 to 1 p.u. at 0 s, and no fault happens. At this time, the operation of the proposed method is shown in Fig. 20.

As shown in Fig. 20(a), the power output of the OWF before a fault will not have an evident impact on the proposed protection method since the current waveforms between both sides are always opposite in normal operation and the fault transient current is always present during the fault inception. In addition, the sudden power output change will not also lead to the malfunction of the proposed protection scheme, as displayed in Fig. 20(b).

G. Impact of the Line Length

Since the role of the MMC-HVDC line is to transmit power over long distances, the OWF is often connected to the MMC nearby such that the tie line between the OWF and the MMC is not too long. When the line length is extended to 40 and 60 km, and metallic AG and BC faults occur at F3, the performance of the proposed protection is shown in Fig. 21.

When the line length is extended to 40 and 60 km, the threshold value will be increased to avoid the influence of the capacitance current, but the calculated Minkowski distance in Fig. 21 can be still higher than the threshold value for the faulty phase, and it will be less than the threshold value for the nonfaulty phase, so the proposed protection can work properly. Compared with Fig. 21(b) and (d), the duration to overtake the threshold value will be slightly shortened, but it is still enough to trigger the protection.

H. Comparisons With Other Methods

Different methods based on transient currents will be compared in this part, such as Euclidean distance, Cosine similarity, and Pearson correlation coefficient. They have been used for the transmission line between the RES power plant and the traditional grid. When they are applied to the OWF-MMC system, the performance of these methods is shown in Fig. 22 for AG faults at F3. For Euclidean distance and Cosine similarity, the used time window is 10 ms, and it is set to 20 ms for Pearson correlation coefficient.

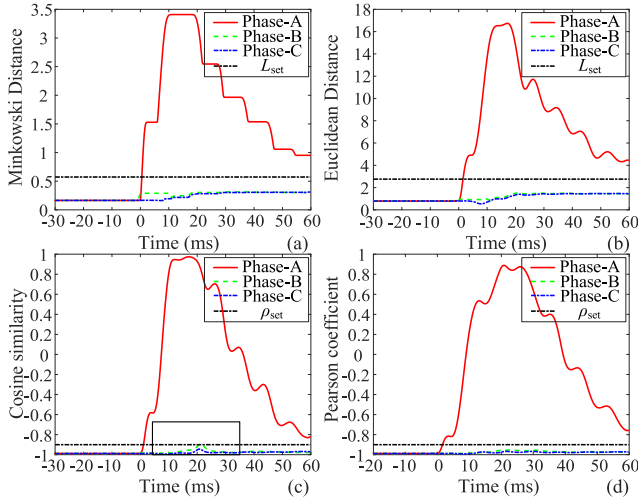


Fig. 22. Performance of different methods. (a) Minkowski distance. (b) Euclidean distance. (c) Cosine similarity. (d) Pearson coefficient.

The threshold value ρ_{set} for Cosine similarity and Pearson coefficient is set to -0.9 [14], [15]. Euclidean distance is a special case of Minkowski distance when p is taken as 2. The operating time is respectively 0.6 ms and 1.6 ms in Fig. 22(a) and (b), so the operating speed for the proposed method is faster than the scheme based on Euclidean distance. This phenomenon is consistent with the analysis in Fig. 14. The method based on the Pearson correlation coefficient and Cosine similarity essentially reflects the phase angle change between the transient currents on both sides and is not sensitive to amplitude changes, so their operating time (1.2 ms and 1.8 ms) are also slower than the proposed scheme, as shown in Fig. 22(c) and (d). In addition, the method based on Cosine similarity also has a poor ability to prevent malfunction of the nonfaulty phase during a fault since the cable capacitance will lead to numerical stability during the transient process. To further analyze the impact of the time window, the operations of different protection methods are displayed in Fig. 23.

As shown in Fig. 23(a), the calculated Minkowski distance for the nonfaulty phase will fluctuate when the time window is set to 5 ms, but it does not overtake the threshold value, so the proposed method can still work properly. However, the operating speed of the Euclidean distance-based pilot protection in Fig. 23(b) is lower than that of the proposed protection scheme. In addition, the fluctuation of the calculated Euclidean distance is more serious. When the Cosine similarity is applied, the calculated value in Fig. 23(c) for the nonfaulty phase will overtake the threshold value during a fault, so the protection will be activated wrongly. Moreover, the calculated Pearson correlation coefficient in Fig. 23(d) will fluctuate within a large scope, and it will be higher than the threshold value even in normal operation, so the protection will be also triggered wrongly. Finally, different indicators are compared in Table III.

VI. RTDS VERIFICATION

A similar OWF-MMC system will be built in RSCAD to verify the operation performance of the proposed method. RSCAD

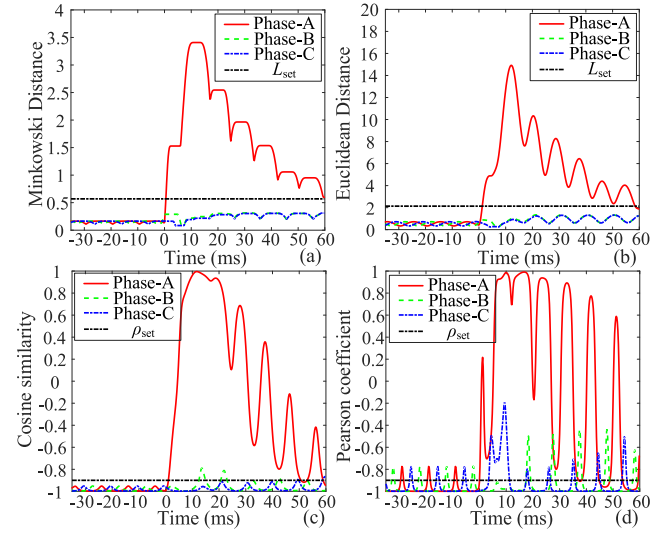


Fig. 23. Performance of different protection methods with a 5 ms time window. (a) Minkowski distance. (b) Euclidean distance. (c) Cosine similarity. (d) Pearson coefficient.

TABLE III
COMPARATIVE ANALYSIS AMONG DIFFERENT METHODS

Approach	Time window	Operating speed	R_g	Noise	Numerical stability
Proposed method	10 ms	Highest	200 Ω	20 dB	High
Euclidean distance	10 ms	Medium	200 Ω	20 dB	High
Cosine similarity	10 ms	Medium	100 Ω	20 dB	Low
Pearson coefficient	20 ms	Lowest	100 Ω	20 dB	High

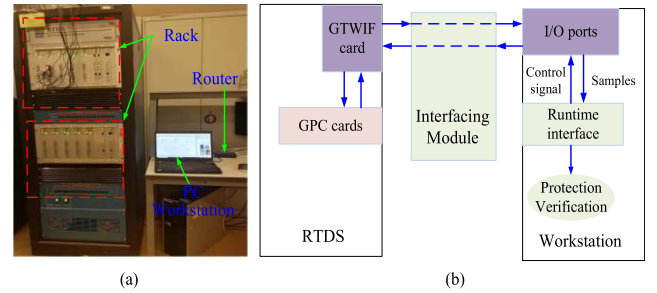


Fig. 24. Established RTDS platform. (a) Hardware devices. (b) Hardware connection diagram.

is a special software environment used to control the RTDS. The rack is the core part of the RTDS, which contains five GPC computing boards and a GTWIF communication card. After the model in RSCAD is compiled, it will be downloaded to the GPC card in the Rack. The real-time simulation data generated in the GPC card will be sent back to the control panel in the workstation through the GTWIF board and network communication equipment, as shown in Fig. 24 [28].

When different fault types occur at F2, the fault current at the fault point can be seen in Fig. 25.

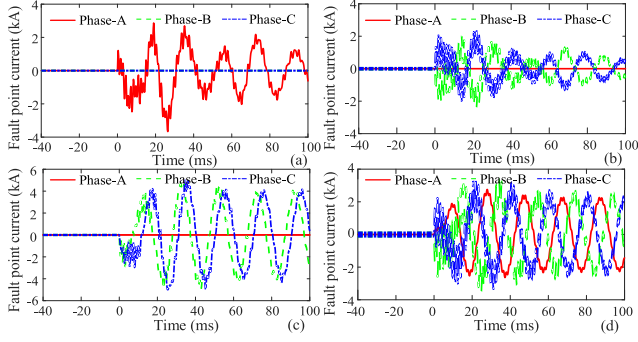


Fig. 25. Fault current at the fault point for different fault types from RTDS testing. (a) AG. (b) BC. (c) BCG. (d) ABCG.

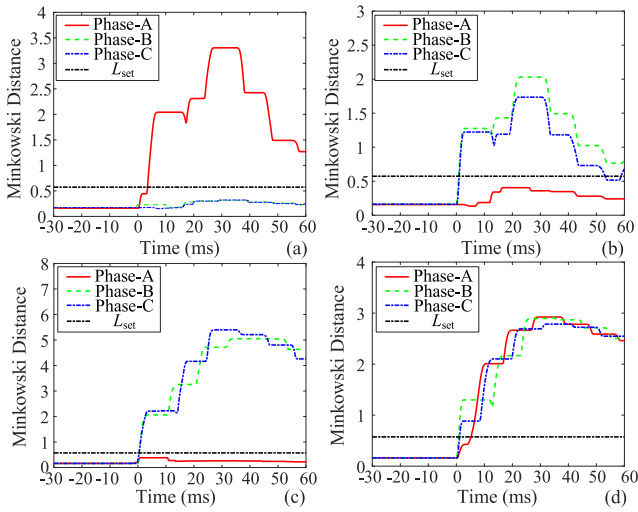


Fig. 26. Performance of the proposed method from RTDS testing. (a) AG. (b) BC. (c) BCG. (d) ABCG.

It can be seen from Fig. 25(a) and (b) that there is an evident fault current during the fault inception (from 0 to 40 ms) for AG and BC faults. After that, the fault current is suppressed, which means the fault features are only apparent at the onset of the fault for the studied system. For BCG faults and ABCG faults in Fig. 25(c) and (d), the fault current is evident and stable during the fault since the fault current will not be blocked by the negative-sequence fault loop. In addition, the current waveform distortion during the fault inception in Fig. 25 is not from noise but from the complex fault transients. In this situation, the performance of the suggested method is displayed in Fig. 26.

As shown in Fig. 26(a) and (b), the computed Minkowski distance will quickly overtake the threshold value for the faulty phases, and this state will last for over two cycles, which is enough to trigger the protection. However, the calculated Minkowski distance will fall below the threshold value after 50 ms in Fig. 26(b) since the fault current at the fault point for BC faults will be suppressed during the steady state, and this phenomenon is also consistent with the PSCAD simulation result in Fig. 17(b). For the nonfaulty phase, the calculated value is always lower than the setting value, so the nonfaulty phase is

blocked reliably. Meanwhile, the calculated Minkowski distance in Fig. 26(c) and (d) will be always higher than the threshold value for BCG faults and ABCG faults due to the stable fault current at the fault point. These phenomena support this method again.

VII. CONCLUSION

It is found that the steady-state fault current is suppressed under single-line ground faults and line-to-line faults for the OWF-MMC system. Therefore, the sensitivity of traditional differential protection is seriously reduced or even fails to operate under high-resistance faults.

To cope with this problem, a new pilot protection based on Minkowski distance is proposed. Minkowski distance can effectively measure the waveform difference between transient currents on both sides, so internal and external faults can be identified effectively. The proposed method does not depend on the power frequency phasor, so it is not affected by the frequency offset during the transient process. At the same time, compared with other methods based on transient current differences, the proposed method has a faster operating speed. In addition, the proposed method is applicable to different FRT strategies and has strong capacities to resist noise interference and fault resistance. PSCAD simulation and RTDS experiments verify the effectiveness of the suggested method.

APPENDIX

TABLE IV
SIMULATION VARIABLES

Components	Variables	Values
Cable line	Zero-sequence impedance	$0.220+j0.928 \Omega/\text{km}$
	Positive-sequence impedance	$0.084+j0.350 \Omega/\text{km}$
	Zero-sequence capacitance	$0.126 \mu\text{F}/\text{km}$
	Positive-sequence capacitance	$0.063 \mu\text{F}/\text{km}$
T1	Rated capacity	300 MVA
	Wire connection	Dyn
	Short circuit impedance	10%
T3	Rated transformation ratio	35/220 kV
	Wiring connection	YND
	Normal capacity	300 MVA
	Rated transformation ratio	220/370 kV
	Short-circuit ratio	10%

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Zhe Yang (Member, IEEE) was born in China in 1994. He received the bachelor's degree in electrical engineering from Northeast Electric Power University, Jilin, China, in 2017, the master's degree in electrical engineering from North China Electric Power University, Beijing, China, in 2020, and the Ph.D. degree in energy technology from AAU Energy at Aalborg University, Aalborg, Denmark, in 2023.

He is currently a Postdoc Researcher with the Hong Kong Polytechnic University, Hong Kong.

His research interests include power system protection and control, machine learning, and deep learning.



Ruijin Zhu received the M.S. degree in electrical engineering from Gansu Agricultural University, Lanzhou, China, in 2014.

He is currently an Associate Professor with China Tibet Agriculture and Animal Husbandry University, Lhasa, China. His current research interests include smart grid and machine learning.



Wenlong Liao (Member, IEEE) received the B.S. degree in electrical engineering from China Agricultural University, Beijing, China, in 2017, the M.S. degree in electrical engineering from Tianjin University, Tianjin, China, in 2020, and the Ph.D. degree in energy technology from Aalborg University, Aalborg, Denmark, in 2023.

He is currently a Postdoc Researcher with the Wind Engineering and Renewable Energy Laboratory, École Polytechnique Fédérale de Lausanne. His current research interests include

smart grids, machine learning, and power system protection.